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Segmented electrode and method

Abstract

A medical lead for implantation includes an electrode assembly having a plurality of electrodes. Each of the plurality of electrodes include a plurality of electrode segments. Each electrode segment has a flexible conductor directly coupled to it. A strut is coupled within each of the plurality of electrodes such that each of the flexible conductors coupled to the plurality of electrode segments are supported between the strut and each of the plurality of electrodes by the strut. An insulator fills gaps between the plurality of electrodes, the strut and the flexible conductors.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This Non-Provisional Patent Application claims the benefit of the filing date of U.S. Provisional Patent Application Ser. No. 62/375,369, filed Aug. 15, 2016, entitled "SEGMENTED ELECTRODE AND METHOD," which is herein incorporated by reference.

BACKGROUND

(1) This disclosure relates to segmented electrodes configured for sensing and/or stimulation within a biological application. In some embodiments, ring electrodes are provided on the distal end of a lead for sensing or stimulation within a human body. The distal end of a lead is placed adjacent tissue that is to be sensed or stimulated and the ring electrodes either transmit or receive energy. In some cases, it is useful to have very discrete locations energized, and accordingly, use only a segment of a ring electrode, rather than the entire ring. Manufacturing discrete electrode segments can be difficult, particularly where multiple electrode segments are desired on a small diameter lead. For these and other reasons, there is a need for the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a side view of a medical lead with segmented electrodes in accordance with one embodiment.
- (2) FIG. 2 illustrates a perspective view of an electrode assembly for a lead in accordance with one embodiment.

- (3) FIG. 3 illustrates an exploded perspective view of a strut and electrode assembly with wiring for a lead in accordance with one embodiment.
- (4) FIG. 4 illustrates a perspective view of an electrode assembly with wiring for a lead in accordance with one embodiment.
- (5) FIG. 5 illustrates a partial perspective view of an electrode assembly with wiring for a lead in accordance with one embodiment.
- (6) FIG. 6A illustrates an end view of an electrode assembly with wiring before centerless grinding in accordance with one embodiment.
- (7) FIG. 6B illustrates an end view of an electrode assembly with wiring after centerless grinding in accordance with one embodiment.
- (8) FIG. 7A illustrates a top view of an electrode assembly with wiring for a lead in a mold cavity in accordance with one embodiment.
- (9) FIG. 7B illustrates a perspective view of an electrode assembly with wiring for a lead after grinding in accordance with one embodiment.
- (10) FIG. 8 illustrates a perspective view of a strut for a lead in accordance with one embodiment.
- (11) FIG. 9 illustrates a perspective view of an electrode assembly with wiring for a lead in accordance with one embodiment.
- (12) FIG. 10 illustrates an end view of an electrode assembly before centerless grinding in accordance with one embodiment.
- (13) FIG. 11 illustrates a partial perspective view of an electrode assembly with wiring for a lead in accordance with one embodiment.
- (14) FIG. 12 illustrates a perspective view of a strut for a lead in accordance with one embodiment.
- (15) FIG. 13 illustrates a perspective view of an electrode from an electrode assembly for a lead in accordance with one embodiment.
- (16) FIG. 14 illustrates a perspective view of an electrode from an electrode assembly for a lead in accordance with one embodiment.
- (17) FIG. 15 illustrates perspective views of wire sleeves for use in an electrode for a lead in accordance with one embodiment.
- (18) FIGS. 16A-16B illustrate side and perspective views of a lead with segmented electrodes in accordance with one embodiment.
- (19) FIGS. 17A-17B illustrate side and perspective views of a lead with segmented electrodes in accordance with one embodiment.
- (20) FIG. 18 illustrates a method of forming a medical lead with segmented electrodes in accordance with one embodiment.

DETAILED DESCRIPTION

(21) In the following Detailed Description, reference is made to the accompanying drawings, which form a part hereof, and in which is shown by way of illustration specific embodiments in which the embodiments may be practiced. In this regard, directional terminology, such as “top,” “bottom,” “front,” “back,” “leading,” “trailing,” etc., is used with reference to the orientation of the Figure(s) being described. Because components of embodiments can be positioned in a number of different orientations, the directional terminology is used for purposes of illustration and is in no way limiting. It is to be understood that other embodiments may be utilized and structural or logical changes may be made without departing from the scope of the embodiments. The following detailed description, therefore, is not to be taken in a limiting sense, and the scope of the embodiments is defined by the appended claims.

(22) FIG. 1 illustrates a side view of a medical lead 10 with segmented electrodes 20, 22, 24, 26 in accordance with one embodiment. In one embodiment, lead 10 includes, adjacent its distal end 12, four electrodes 20, 22, 24, 26, each of which is segmented such that each has a plurality of individually accessible electrode segments. Specifically, first electrode 20 includes multiple electrode segments 20a; second electrode 22 includes multiple electrode segments 22a; third

electrode **24** includes multiple electrode segments **24a**; and fourth electrode **26** includes multiple electrode segments **240a**. In one various embodiments, there are two, three, four or five electrode segments **20a**, **22a**, **24a**, **26a**, for each of electrodes **20**, **22**, **24**, **26**.

(23) In operation, lead **10** may be configured for use within a human body. Once within a human body, each of electrode segments **20a**, **22a**, **24a**, **26a** may be used for directional stimulation or for positional feedback sensing. Rather than using a single ring electrode that spans the entire 360° circumference of the lead, lead **10** includes electrode segments **20a**, **22a**, **24a**, **26a**, which only span a portion of the circumference of lead **10** (for example, 180°, 90° degrees or less), such that directional stimulation or positional feedback sensing can be much more precisely controlled relative to a given target within the human body.

(24) Furthermore, lead **10** in accordance with embodiments described herein, allow for the manufacture of leads having increased density of electrode segments. Increased density of electrode segments is useful in a variety of applications. For example, lead **10** can be used in deep brain stimulation (DBS), in which lead **10** delivers electrical pulses into one or several specific sites within the brain of a patient to treat various neurological disorders, such as chronic pain, tremors, Parkinson's disease, dystonia, epilepsy, depression, obsessive-compulsive disorder, and other disorders. In other applications, lead **10** may be configured for spinal cord stimulation, peripheral nerve stimulation, dorsal root stimulation, cortical stimulation, ablation therapies, cardiac rhythm management leads, various catheter configurations for sensing, and various other therapies where directional sensing or stimulation are needed.

(25) In one embodiment, the manufacture of lead **10** begins with electrode assembly **30**, such as illustrated in FIG. 2. Electrode assembly **30** includes first electrode **20**, second electrode **22**, third electrode **24**, and fourth electrode **26**. Each electrode is coupled to at least on other electrode with coupling segments **32**. Each electrode **20**, **22**, **24**, **26** is provided with a plurality of grooves **20b**, **22b**, **24b**, **26b**, respectively, on the outer periphery of the electrode. The number of grooves provided in each electrode **20**, **22**, **24**, **26** corresponds to the number of electrode segments **20a**, **22a**, **24a**, **26a** that will be provided for each electrode. A flexible conductor (not illustrated in FIG. 2) will be coupled within each groove **20b**, **22b**, **24b**, **26b**, thereby provided independent electrical access to each electrode segment **20a**, **22a**, **24a**, **26a**.

(26) Electrode assembly **30** can be formed in a variety of way consistent with the embodiments. For example, electrode assembly **30** can be formed by machining, metal injection molding, 3-D printing and/or metal screen printing. Fabrication of electrode assembly **30** in this way simplifies the manufacturing process and provides precise and consistent electrodes. Forming electrode assembly **30** as a single part, with electrodes **20**, **22**, **24**, **26** coupled together with coupling segments **32** also ensures that precise spacing between adjacent electrodes is maintained in the final lead **10**, which can be important in many applications.

(27) Electrode assembly **30** is illustrated in FIG. 3 coupled to a plurality of flexible conductors **40**. Strut **50** is illustrated exploded away from electrode assembly **30**. In one embodiment, a flexible conductor **42**, **44**, **46**, **48** is placed in each groove **20b**, **22b**, **24b**, **26b** of each electrode **20**, **22**, **24**, **26**. For example, where each electrode **20**, **22**, **24**, **26** includes four grooves **20b**, **22b**, **24b**, **26b**, the plurality of flexible conductors **40** includes a total of 16 flexible conductors. Once the plurality of flexible conductors **40** are coupled within the respective grooves **20b**, **22b**, **24b**, **26b**, strut **50** is placed within electrode assembly **30**, thereby supporting the plurality of flexible conductors **40** between strut **50** and electrode assembly **30**.

(28) FIG. 4 illustrates strut **50** assembled within electrode assembly **30**. In one embodiment, strut **50** is provided with a stop **53** and a plurality of helical channels **52**. During assembly, strut **50** is inserted into electrode assembly **30** up to stop **53**, such that stop **53** is adjacent electrode **20** as illustrated in FIG. 4. Stop **53** ensures proper axial alignment of strut **50** within electrode assembly **30**, such that the length of strut **50** extends within each of electrodes **20**, **22**, **24**, **26** and supports the plurality of flexible conductors **40** therein. In one embodiment, strut **50** is also provided with a

center lumen **51**, illustrated in FIG. 5, where stop **53** is removed to reveal center lumen **51** and the plurality of helical channels **52** in the end view.

(29) In one embodiment where 16 flexible conductors make up the plurality of flexible conductors **40**, four helical channels **52** are provided in strut **50**. In this way, each flexible conductor **42**, **44**, **46**, **48** that is coupled to a groove **20b**, **22b**, **24b**, **26b** is then placed into a helical channel **52** of strut **50**. Each channel **52** is configured to accommodate up to four flexible conductors. For example, each of the four flexible conductors **42** coupled within the four grooves **20b** of first electrode **20** are each placed in one of the four channels of strut **50**. Then, moving down the length of electrode assembly **30**, four more flexible conductors **44** that are coupled within the four grooves **22b** of second electrode **22** are each placed in one of the four channels of strut **50** adjacent the flexible conductors from second electrode **22**. Again, moving down the length of electrode assembly **30**, four more flexible conductors **46** that are coupled within the four grooves **24b** of third electrode **24** are each placed in one of the four channels of strut **50** adjacent the flexible conductors from first and second electrodes **20** and **22**. Finally, once flexible conductors **48** are coupled to fourth electrode **24**, there will be four flexible conductors in each of the four channels **52** of strut **50**. As such, each of the flexible conductors are well supported during subsequent molding processes and during use of lead **10**.

(30) In one embodiment, the plurality of flexible conductors **40** are secured to electrodes **20**, **22**, **24**, **26** using wedges **20c**, **22c**, **24c**, **26c**, as illustrated in FIG. 5 (not all wedges are shown for simplification of the figure). Once a flexible conductor is placed in a groove, such as groove **20b** illustrated in FIG. 5, wedge **20c** is then placed within groove **20b** over the flexible conductor **42**, thereby securing flexible conductor **42**. In one embodiment, wedge **20c** is of the same material as electrode **20** and can be resistance or laser welded into place.

(31) Because the plurality of flexible conductors **40** are supported by strut **50** and secured to electrodes **20**, **22**, **24**, **26**, flexible conductors are wires or cables that are flexible and easily threaded up into electrodes **20**, **22**, **24**, **26**, thereby simplifying the manufacturing process. In some previous designs, hypotubes, or stiff metal tubes were welded to the inner periphery of electrodes. Using the stiff metal tubes complicates both the assembly and the welding process.

(32) FIG. 6A illustrates an end view of first electrode **20** with wedges **20c** secured within each of four grooves **20b** on the outer periphery of electrode **20**. Flexible conductors **42** are illustrated firmly secured under each of wedges **20c**. Strut **50** is also illustrated within electrode **20**. For illustration purposes, stop **53** is removed so that four helical channels **52** are illustrated, each one for supporting one of the four flexible conductors **42** secured within grooves **20b**. Strut **50** supporting the plurality of flexible conductors **40** in this way minimizes strain on the flexible conductors and secures during subsequent molding processes, as will be further discussed.

(33) Once each of flexible conductors **42** are secured within grooves **20b**, electrode **20** is then ground from its outer periphery inward to grind line **60** in order to define electrode segments **20a**. In one embodiment, a centerless grinding process is used to grind down to grind line **60**. In its original configuration as part of electrode assembly **30**, electrode **20** includes openings **62** along its inner periphery. As electrode **20** is ground down to grind line **60**, openings **62** are then exposed in the post-grind outer periphery of electrode **20**, thereby defining electrode segments **20a**.

(34) FIG. 6B illustrates electrode **20** after the grinding process. For each of electrode segments **20a** defined between openings **62**, flexible conductors **42** are embedded and secured between the outer periphery **21** and inner periphery **19** of electrode segments **20a**. Furthermore, each electrode segment **20a** is electrically isolated from each other segment by opening **62** and can be individually accessed via each of the four flexible conductors **42**. FIGS. 6A and 6B illustrate the processing of electrode **20** before and after grinding, but processing the other electrodes **22**, **24** and **26** is directly analogous. The number of openings **62** provided in the electrodes **20**, **22**, **24**, **26** of electrode assembly **30** determines the number of electrode segments that will be defined after grinding. Although four openings are illustrated, two, three, four or five openings can be provided to

respectively define two, three, four or five electrode segments **20a**. The number of flexible conductors **42** provided also matches the number of electrode segments **20a** so that each can be independently accessed.

(35) In one embodiment, the centerless grinding process can be done before the electrode assembly is molded, and in another embodiment, the centerless grinding process is done after the assembly is molded. FIG. 7A illustrates the later embodiment, where strut **50** assembled within electrode assembly **30** and coupled to the plurality of flexible conductors **40** (such as illustrated in FIG. 4), is placed within injection mold **75**. Injection mold **75** includes mold cavity **75b**, in which electrode assembly **30** is placed, and mold gates **75a**. Once mold cavity **75b** is closed against its mirror image cavity (not illustrated), molding material, such as thermoplastic or elastomer insulation, is flowed into cavity **75b** via mold gates **75a**. The mold material fills all spaces within electrode assembly **30**, including filing around the combination of strut **50** and the plurality of flexible conductors **40** between each of the electrodes **20**, **22**, **24**, **26**, and also filing any small spaces within each of electrodes **20**, **22**, **24**, **26**. Because the plurality of flexible conductors **40** are supported within strut **50**, they are not significantly disturbed by the force with which the mold material enters cavity **75b**, and are instead supported and held in place between strut **50** and within each of electrodes **20**, **22**, **24**, **26**.

(36) After the mold material solidifies, the entire assembly is removed and centerless ground to form lead **10**, as illustrated in FIG. 1. Molding material fills the spaces between each of electrode segments **20a**, **22a**, **24a**, **26a** such that each is electrically isolated from each other by the insulative molding material. Each of the plurality of flexible conductors **40** are surrounded and firmly secured with the molding material.

(37) When an elastomer is used as the mold material, lead **10** has some flexibility, which is useful in some applications, such as where lead **10** is incorporated into a catheter configured to navigate the tortuous vasculature of a human body. In such case, the helical configuration of the strut **50**, with the plurality of flexible conductors **40** following the helical configuration, allows relief to the flexible conductors as the lead **10** bends as it moves through the vasculature. If the flexible conductors are all aligned parallel, rather than twisted around the helical configuration, too much bending can cause too much strain on the flexible conductors.

(38) FIG. 7B illustrates further processing of lead **10** in accordance with one embodiment. As illustrated, each of electrodes **20**, **22**, **24**, **26** have been ground to define a plurality of electrode segments **20a**, **22a**, **24a**, **26a** (four segments in the illustrated example, although only three are visible in the view). Four each of flexible conductors **42**, **44**, **46**, **48** are respectively embedded in electrode segments **20a**, **22a**, **24a**, **26a** such that the plurality of flexible conductors **40** include a total of 16. Each of the plurality of flexible conductors **40** are secured within the helical channels **52** of strut **50**. In order to further illustrate the internal portions of lead **10**, all molding material has been removed from the illustration.

(39) With the centerless grinding process, in addition to removing an outer periphery of each of electrodes **20**, **22**, **24**, **26**, the grinding also removes each of the coupling segments **32** between the electrodes. The axial location of each of the electrodes relative to each other, however, is maintained by strut **50**, which is secured within electrode assembly **30** before the grinding process. Strut **50** within electrode assembly **30** is configured to extend through each of electrodes **20**, **22**, **24**, **26** and thereby constrain the axial distance and axial orientation of each of the electrodes **20**, **22**, **24**, **26** even after the coupling segments **32** have been ground away.

(40) In one embodiment, lead **10** also includes marker bands **72** and **74**. Marker bands are useful in locating lead **10** once inserted in a human body via magnetic imaging. Accordingly, by locating the marker bands, the axial location and orientation of the electrode segments **20a**, **22a**, **24a**, **26a** can be determined.

(41) FIG. 8 illustrates strut **80** in accordance with one embodiment. Strut **80** is analogous to strut **50** illustrated in FIG. 4. In one embodiment, strut **80** includes a stop **90**, a center lumen **81**, and four

ribs **82, 84, 86, 88**, thereby defining four channels **83, 85, 87, 89**, each channel defined between two adjacent ribs. In one embodiment, multiple flexible conductors from a plurality of flexible conductors **120** can be placed in each channel **83, 85, 87, 89**, in order to support and maintain the position of the flexible conductor, and isolate flexible conductors in one channel from those in other channels.

(42) FIG. **9** illustrates an electrode assembly **110** coupled over strut **80** with a plurality of flexible conductors **120** coupled thereto. As with the electrode assembly **30** previously described, electrode assembly **110** includes first, second, third and fourth electrodes **100, 102, 104, 106**, which are coupled over strut **80**. In one embodiment, each of electrodes **100, 102, 104, 106** will be segmented with a centerless grinding process to form four electrode segments each. In other embodiments, two, three or five segments are possible for each electrode. In the illustrated embodiment with four electrode segments, each electrode **100, 102, 104, 106** couple with four flexible conductors—one connected to each of its segments, as will be more fully described. As such, strut **50** accommodates four flexible conductors in each of its four channels **83, 85, 87, 89** between each of its four ribs **82, 84, 86, 88**, thereby supporting the plurality of flexible conductors **120** between strut **80** and electrodes **100, 102, 104, 106**.

(43) During assembly, strut **80** is inserted into electrode assembly **110** up to stop **90**, such that stop **90** is adjacent electrode **106** as illustrated in FIG. **9**. Stop **90** ensures proper axial alignment of strut **80** within electrode assembly **110**, such that the length of strut **80** extends within each of electrodes **100, 102, 104, 106** and supports the plurality of flexible conductors **120** therein, and separates them into channels **83, 85, 87, 89**. In one embodiment, strut **80** is also provided with a center lumen **81**, illustrated in FIG. **8**.

(44) FIG. **10** illustrates an end view of the electrode assembly **110** and strut **80** from FIG. **9**. Electrode **100** includes openings **92**. In one embodiment, four openings **92** are provided so that four electrode segments **110a** are defined by the grinding process. As previously described relative to electrode assembly **30** and strut **50**, electrode assembly **110** and strut **80** are ground from the periphery down to grind line **127**. This grinding exposes openings **92**, thereby defining electrode segments **100a**, each of which are electrically isolated from each other by openings **92**. Each on openings **92** are filled with molding material by the molding process described above relative to electrode assembly **30**.

(45) Electrode **100** also includes notches **115**. Notches **115** provide a flexible conductor retaining feature, analogous to that provided by grooves **20b, 22b, 24b, 26b** in electrode assembly **30**, which facilitates attaching one flexible conductor of the plurality of flexible conductors **120** to each electrode segment **100a**.

(46) FIG. **11** illustrates a more detailed view of a portion of electrode assembly **110**, and connections between the plurality of flexible conductors **120** and electrode segments **106a**. The electrode assembly **110** illustrated is after a grinding process, such that electrode segments **106a** have been defined, and molding material is either not yet added or has been removed for illustration purposes. Each electrode segment **106a** includes a shoulder **125**. Notches **115** are provided in shoulder **125** for each electrode segment **106a**. Accordingly, four flexible conductors **121** of the plurality of flexible conductors **120** are attached to the four notches **115**, one for each of the four electrode segments **106a**.

(47) In one embodiment, each flexible conductor **121** includes a bend at its end **130**, such that it is more readily inserted into notch **115** and can easily be welded into place. In one embodiment, the bend at end **130** is approximately 90 degrees. Flexible conductors **121** are fed through channels **83, 85, 87, 89** and up to one of the electrodes. The bend at the flexible conductor end **130** is then placed into notch **115** and welded there, for example, resistance or laser welded. As illustrated in FIG. **11**, each flexible conductor **121** coupling to each electrode segment **106a** is in a separate channels **83, 85, 87, 89**, and thus, separated from each other by a rib **82, 84, 86, 88**. This affords management of the plurality of flexible conductors **120** and streamlines the assembly process and

avoids confusing flexible conductors **121**. Furthermore, each of the flexible conductors are well supported during subsequent molding processes and during use of lead **10**.

(48) Once all flexible conductors are attached to electrode segments **100a**, **102a**, **104a**, **106**, the assembly is placed in a mold, such as mold **75** in FIG. 7A and overmolded as previously described. After the mold material solidifies, the entire assembly is removed and centerless ground to form lead **10**, as illustrated in FIG. 1. As with previous designs, the centerless grinding process can be done before the electrode assembly is molded, or the centerless grinding process may be done after the assembly is molded.

(49) FIG. 12 illustrates strut **150** in accordance with one embodiment. Strut **150** is analogous to struts **50** and **80** previously described and can be used in any of the previous embodiments. In one embodiment, strut **150** further includes positioning arms **162**, **164**, **166**, **168**. Each positioning arm is configured with a catch **162a**, **164a**, **166a**, **168a**. Strut **150** may be used as strut **50** or **80** described previously, but also provide additional positioning of electrodes. For example, any of electrode **20**, **22**, **24**, **26** in FIG. 2 can be fed over strut **150** and snapped into place via positioning arms **162**, **164**, **166**, **168** and catches **162a**, **164a**, **166a**, **168a**. First electrode **20** can be moved over each of the positioning arms **162**, **164**, **166**, **168** and snapped into place in fourth positioning arm **168**. Catch **168a** will then engage a feature in electrode **20** such that it cannot be moved back in the direction from which it was assembled. The next three electrodes **22**, **24**, **26** can be assembled over the three remaining positioning arms **162**, **164**, **166** in the same way.

(50) In preventing back movement of each electrode, positioning arms **162**, **164**, **166**, **168** and catches **162a**, **164a**, **166a**, **168a** ensure that each electrode is precisely positioned along the length of strut **150** correctly and ensures that the relative distance between each electrode is precisely controlled. This is important in many applications where positioning of the electrode segment relative to tissue in the human body must be very precise. Also, catches **162a**, **164a**, **166a**, **168a** preventing backward movement along the length of strut **150** is useful during the injection molding process. Even where significant flow forces are generated by molten molding material moving into the mold cavity, catches **162a**, **164a**, **166a**, **168a** prevent electrodes from being moved axially by these forces.

(51) FIG. 13 illustrates an electrode **200** in accordance with one embodiment. Electrode **200** is analogous to those previously described and can be used in any of the previous embodiments. Electrode **200** includes grooves **202** and openings **204**. In the Figure, four grooves **202** and openings **204** are illustrated, but other amounts, for example, two, three or five, are possible. In each groove **202**, sleeve **210** is provided over a flexible conductor. Sleeve **210** helps facilitate a good connection between the plurality of flexible conductors **40**, **120** and electrode **200**.

(52) Although only a single sleeve **210** is illustrated in FIG. 13 for ease of illustration, in assembly of lead **10** a sleeve **210** is inserted in each of the four grooves **202**. In one embodiment, each flexible conductor is inserted into a sleeve **210** and then a crimp **210a** is made on sleeve **210** to secure the flexible conductor within sleeve **210**. Once the flexible conductor is secure within sleeve **210**, sleeve **210** is placed within groove **202** and can be welded into place. In one embodiment, sleeve **210** is welded at each of its ends.

(53) FIG. 14 illustrates an electrode **220** in accordance with one embodiment. Electrode **220** is analogous to those previously described and can be used in any of the previous embodiments. Electrode **220** includes grooves **222** and openings **224**. In the Figure, four grooves **222** and openings **224** are illustrated, but other amounts, for example, two, three or five, are possible. In each groove **222**, sleeve **242** is provided over a flexible conductor. Sleeve **242** helps facilitate a good connection between the plurality of flexible conductors **40**, **120** and electrode **220**.

(54) Although only a single sleeve **242** is illustrated in FIG. 14 for ease of illustration, in assembly of lead **10** a sleeve **242** is inserted in each of the four grooves **222**. In one embodiment, each flexible conductor is inserted into a sleeve **242** and then a crimp is made on sleeve **242** to secure the flexible conductor within sleeve **242**. Once the flexible conductor is secure within sleeve **242**,

sleeve **210** is slid into groove **222** and can be welded into place. In one embodiment, sleeve **242** is welded at its end.

(55) FIG. **15** illustrates various embodiments of sleeves **210**, **240**, **242**, **244** over flexible conductors **210a**, **240a**, **242a**, **244a** for use in the electrodes previously described. As illustrated, sleeves **210**, **240**, **242**, **244** for use in electrodes, such as electrodes **200**, **220** illustrated in FIGS. **13-14**, can be of various configurations. Sleeve **210** is generally cylindrical and over flexible conductor **210a**, similar to that illustrated in groove **202** of FIG. **13**. Sleeve **242** is also generally cylindrical with a longitudinal slot along its length. Sleeves **240** and **244** are generally rectangular, sleeve **244** being closed and sleeve **240** has an open slot. Each of sleeves **210**, **240**, **242**, **244** can include a crimp over flexible conductors **210a**, **240a**, **242a**, **244a** to ensure the respective sleeve tightly retains the flexible conductor. Each sleeve can then be welded into the respective groove of an electrode.

(56) FIGS. **16A** and **16B** illustrate side and perspective views of lead **310** in accordance with one embodiment. Lead **310** includes four electrodes **320**, **322**, **324**, **326** out toward its distal tip **312**. Each of the four electrodes **320**, **322**, **324**, **326** are segmented such that each has independently accessible electrode segments **320a**, **322a**, **324a**, **326a**. In the illustrated embodiment, each of the electrode segments **320a**, **322a**, **324a**, **326a** are radially aligned. For example, each electrode segment **320a** of electrode **320** is aligned with each electrode segment **322a** (and the other segments **324a**, **326a**), such that their edges lie on a continuous straight line along the length of lead **310**. This configuration may be useful in certain application for ensuring proper location of the electrode segments relative to tissue that is being stimulated or sensed.

(57) FIGS. **17A** and **17B** illustrate side and perspective views of lead **410** in accordance with one embodiment. Lead **410** includes four electrodes **420**, **422**, **424**, **426** out toward its distal tip **412**. Each of the four electrodes **420**, **422**, **424**, **426** are segmented such that each has independently accessible electrode segments **420a**, **422a**, **424a**, **426a**. In the illustrated embodiment, each of the electrode segments **420a**, **422a**, **424a**, **426a** are radially offset. For example, each electrode segment **420a** of electrode **420** is radially rotated slightly relative to each electrode segment **422a** (and the other segments **424a**, **426a**), such that none of their edges lie in a straight line along the length of lead **310**. This configuration may be useful in certain application for accessing certain locations of the electrode segments relative to tissue that is being stimulated or sensed.

(58) FIG. **18** illustrates a method of forming a segmented electrode lead, such as leads **10**, **310** and **410** in accordance with one embodiment. First, at step **502**, an electrode assembly, such as previously described is formed. The electrode assembly may be formed by any of a variety of processing, including machining, metal injection molding, 3-D printing and/or metal screen printing. The formed electrode assembly has a plurality of electrodes, for example, two, three, four or five. In one embodiment, each of the electrodes in the electrode assembly are each configured with opening and grooves. The grooves are useful in coupling conductors and the openings will define electrode segments.

(59) At step **504**, a plurality of flexible conductors are attached to each of the plurality of electrodes. In one embodiment, a single flexible conductor is attached to each portion of each electrode that will be formed into an electrode segment. For example, if an electrode will be formed into three electrode segments, then three flexible conductors will be attached to that electrode, one flexible conductors connected at each of the segments. In one embodiment, a single flexible is coupled in each groove provided in the electrodes.

(60) At step **506**, the plurality of flexible conductors are arranged relative to a strut. In some embodiments, the flexible conductors are helically wrapped in channels of the strut. In other embodiments, the flexible conductors are sorted into separate channels of the strut. Once the flexible conductors are arranged, the strut is inserted into the electrode assembly, such that the strut is within each of the electrodes in the assembly.

(61) At step **508**, the combination of the flexible conductors, the strut and the electrode assembly

are placed into a mold cavity, and molding material is injected into the cavity. Gaps between the flexible conductors, the strut and the electrode assembly are filled with the molding material, which is allowed to solidify.

(62) At step **510**, the molded electrode assembly is ground inward from its outer periphery. In one embodiment, a centerless grinding process is used to grind the molded electrode assembly down to a grind line that exposed the openings that are formed in each electrode. In so doing, each electrode is segmented, such that a plurality of electrode segments are defined wherein each is electrically isolated from each other.

(63) The various described steps are not necessarily required in a particular order. For example, a centerless grinding process may be applied to the combination of the flexible conductors, the strut and the electrode assembly before the combination is placed into a mold cavity. The injection molding process can then be the last step in producing the finished lead.

(64) Although specific embodiments have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific embodiments shown and described without departing from the scope. This application is intended to cover any adaptations or variations of the specific embodiments discussed herein. Therefore, it is intended that these embodiments be limited only by the claims and the equivalents thereof.

Claims

1. A medical lead for implantation comprising: an electrode assembly comprising a plurality of electrodes; wherein each of the plurality of electrodes comprise a plurality of electrode segments and wherein each electrode segment has a groove and a flexible conductor directly coupled in the groove; a strut having channels and coupled within the plurality of electrodes such that each of the flexible conductors coupled to the plurality of electrode segments are supported by the channels of the strut; and an insulator filling gaps between the plurality of electrodes, the strut and the flexible conductors; wherein at least one electrode segment comprises a wedge configured to hold the flexible conductor against the groove of the at least one electrode segment thereby securing the flexible conductor between the wedge and the at least one electrode segment, wherein centerless grinding of the electrode assembly exposes openings between the plurality of electrode segments and exposes at least a portion of the wedge on an outer surface of the at least one electrode segment; and wherein the wedge and the at least one electrode segment are of the same material.
2. The medical lead of claim 1, wherein each electrode segment comprises an inner and an outer periphery and wherein no part of the flexible conductor is attached to either the inner or outer periphery.
3. The medical lead of claim 1, wherein the lead contains no hypotubes and wherein the strut is configured to extend through each one of the plurality of electrodes.
4. The medical lead of claim 1, wherein the plurality of electrode segments comprise a feature configured to receive the flexible conductor and to secure the flexible conductor to the electrode segment.
5. The medical lead of claim 1, wherein the strut comprises a stop configured to engage the electrode assembly and ensure proper relative positioning of the strut and electrode assembly.
6. The medical lead of claim 1, wherein the electrode assembly comprises between two and four electrodes and wherein each electrode comprises between two and five electrode segments.
7. The medical lead of claim 1, wherein the strut further comprises positioning arms configured to receive and space the electrodes relative to each other.
8. The medical lead of one of claim 7, wherein the positioning arms are further configured with stops to prevent movement of the electrodes in at least one axial direction.
9. The medical lead of claim 1 further comprising sleeves configured over the flexible conductors

and within the grooves of the electrodes.

10. The medical lead of claim 9, wherein the sleeves are crimped over the flexible conductors and are rectangular or cylindrical shaped.

11. The medical lead of claim 1, wherein the segmented electrodes are one of axially aligned and axially offset.

12. A method of manufacturing a medical lead comprising: manufacturing an electrode assembly comprising a plurality of electrodes; forming a plurality of grooves in each of the plurality of electrodes; coupling a flexible conductor directly within each groove of the plurality of grooves in each of the plurality of electrodes, including securing a wedge over the flexible conductor and against each groove thereby securing the flexible conductor between the wedge and each groove; configure the flexible conductors to a strut and attach the strut to the plurality of electrodes; and fill gaps between the plurality of electrodes, the strut and the flexible conductors with an insulator material; and forming a plurality of electrode segments comprising centerless grinding the plurality of electrodes down to a grind line thereby exposing openings between adjacent electrode segments and exposing at least a portion of the wedge and wherein the wedge and the at least one electrode segment are of the same material.

13. The method of claim 12 further comprising inserting the strut up to a stop such that the strut extends through each one of the plurality of electrodes.

14. The method of claim 12 further comprising feeding the flexible conductors through a plurality of channels in the strut, wherein the electrodes are spaced relative to each on the strut using positioning arms on the strut and further comprising sliding sleeves over the flexible conductors and within the grooves of the electrodes.
